CPE 381 Classwork 13: Fourier Transform

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: October 21, 2026 (In-Class)

1. If $x(t) \leftrightarrow X(\Omega)$ is the Fourier transform pair, what is the Fourier transform of $x(t-\tau)$ where τ is the time shift parameter? **(2 Point)**

$x(t-\tau) \longleftrightarrow X(\Omega)\, e^{-j\Omega\tau}$

2. What is Fourier transform of the impulse function $\delta(t)$? **(1 Point)**

$\delta(t) \longleftrightarrow 1$

3. If $x(t) \leftrightarrow X(\Omega)$ is the Fourier transform pair, what is the Fourier transform of $\frac{d}{dt}x(t)$? **(2 Point)**

$\frac{d}{dt}(x(t)) \longleftrightarrow (j\Omega)\, X(\Omega)$

4. Find the Fourier transform of $x(t) = u(t-4) + u(t-5)$ by employing derivative and time-shift properties of Fourier transform. **(15 Points)**

$x(t) = u(t-4) + u(t-5)$

Take derivative both sides

$\frac{d}{dt}(x(t)) = \delta(t-4) + \delta(t-5)$

Take Fourier Transform both sides

$$j\Omega X(\Omega) = 1 \cdot e^{-j\Omega \cdot 4} + 1 \cdot e^{-j\Omega \cdot 5}$$

$$\Rightarrow \quad X(\Omega) = \frac{e^{-j\Omega \cdot 4} + e^{-j\Omega \cdot 5}}{j\Omega}$$